

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE 2500: Engineering Probability, Spring 2026

Homework #8: due Thursday, Apr. 16th, at 2:00 PM.

Show all work for full credit!

Submit your work as a single PDF on Gradescope, labeling each problem number with a page.

1. (35 points) Suppose X is a uniform random variable on $[0, 5]$ and Y is an exponential random variable with parameter $\lambda = 4$. X and Y are independent. Let $Z = X + Y$.

- (a) (10 points) Compute $E(Z)$.
- (b) (10 points) Compute $\text{Var}(Z)$.
- (c) (15 points) Compute $f_Z(z)$. Be sure to specify the PDF for all values of z ; there are three cases. (This is called a convolution integral; drawing a picture may help.)

2. (25 points) The weight of a Smiski in pounds has the PDF given by

$$f_X(x) = \begin{cases} \frac{3}{2}x - \frac{3}{4}x^2 & x \in [0, 2] \\ 0 & \text{otherwise} \end{cases}$$

- (a) (10 points) Use the Chebyshev inequality (the version in Equation 7.20 on p. 366) to find a lower bound on the probability that a collection of 100 independent Smiski weighs between 85 and 115 pounds.
- (b) (10 points) Now use the Central Limit Theorem to estimate the probability that a collection of 100 independent Smiski weighs between 85 and 115 pounds.
- (c) (5 points) How does this compare to the Chebyshev bound? What explains the difference? Which is more accurate?

3. (40 points) The Artemis II crew has a lot of science problems to address on every day of their mission.

- (a) (13 points) Suppose the time it takes Christina to fix the space toilet when it breaks is uniformly distributed in the interval $[4, 6]$ minutes. Use the Central Limit Theorem to estimate the probability that she takes more than 3 cumulative hours to fix the toilet 35 times.
- (b) (13 points) The number of camera SD cards that Reid goes through every day is modeled as a Poisson random variable with mean 4. Use the Central Limit Theorem to estimate how many SD cards NASA should provide for the 10-day mission to be 98% sure he won't run out of cards to take pictures. (Obviously NASA can't provide a fraction of an SD card.)
- (c) (14 points) The amount of data required to transmit a status update to Houston over the ship's optical communication link is modeled as a Gaussian random variable with unknown mean μ MB and standard deviation 20 MB. Each update is independent of every other update. Victor sends 31 status updates to Houston and finds he has communicated 2026 MB. Compute the 95% confidence interval for the mean μ .